

3M Trading Strategy

Complete Context Document — For Claude Code Sessions

Overview

This is a multi-timeframe futures trading strategy called **3M ELITE** built around market structure breaks, supply/demand zones, and a proprietary stage system. The strategy is being automated via TradingView Pine Script → TradersPost → Tradovate (master demo account) → trade copier to multiple prop firm accounts.

Timeframe Presets (1min preset assumed throughout)

Preset	T0	T1	T2	T3	T4	T5
1min	1min	15min	1h	4h	8h	Daily
30sec	30sec	8min	30min	2h	4h	12h

Market Structure Logic

The indicator tracks two structural levels: **bH** (upper range) and **bL** (lower range).

- Bullish break (mkU): price closes or opens above bH AND above previous high (pH). Increments nU. Resets nDn to 0.
- Bearish break (mkDn): price closes or opens below bL AND below previous low (pL). Increments nDn. Resets nU to 0.
- Models: Accumulation, Reaccumulation (bullish), Distribution, Redistribution (bearish)

Supply & Demand Zone Logic

Core concept: Every zone represents exactly ONE candle. The zone box is drawn from that candle's wick low to its wick high (full range).

Zone candle vs engulfing candle

- Zone candle: the candle whose range becomes the zone box
- Engulfing candle: candle immediately to the RIGHT of the zone candle (one bar closer to current bar)
- Demand engulfing: close of engulfing candle must be ABOVE zone candle's HIGH (wick to wick)
- Supply engulfing: close of engulfing candle must be BELOW zone candle's LOW (wick to wick)

Search logic

- mkU (bullish break) → search for DEMAND zone: find candle with LOWEST LOW in range [IDB to bkB-1]
- mkDn (bearish break) → search for SUPPLY zone: find candle with HIGHEST HIGH in range [IUB to bkB-1]

Validity checks (all must pass)

1. Engulfing: $\text{close}[\text{cOff}-1] > \text{high}[\text{cOff}]$ for demand; $\text{close}[\text{cOff}-1] < \text{low}[\text{cOff}]$ for supply
2. Mitigation: no candle after engulfing wicks into zone range. Wick touch = mitigated. $(\text{low}[k] < \text{zone_high AND high}[k] > \text{zone_low})$

- 3. One candle rule: fewer than 2 candles after engulfing trade beyond engulfing extreme

Zone mitigation (ongoing)

- Zone becomes mitigated: price taps it AND then delivers to next structural high/low
- Just tapping alone does NOT mitigate it

Full Strategy Entry Checklist (1min preset, longs)

4h AND 8h (at least one must be bullish):

- Model = Accumulation or Reaccumulation
- $nU \leq 6$ consecutive breaks

1h:

- Model = Accumulation or Reaccumulation
- $nU \leq 6$ consecutive breaks
- Deepest unmitigated demand zone on 1h must have been tapped

15min:

- Model = Accumulation or Reaccumulation
- $nU \leq 6$ consecutive breaks
- Deepest unmitigated demand zone on 1h OR 15min tapped
- Type 1 OR Type 2 validation after zone tap
- 15mpl (15min protected low) set and not invalidated
- Price below KC upper band (length 20, mult 1.25, ATR 14)
- MAs not fully against bias: NOT ($SMA200 > SMA50 > SMA20$)
- Either 20 SMA or 50 SMA sloping upward or flat

1min:

- In valid stage (see Stage System)
- $nU \leq 4$ consecutive breaks
- MAs not fully against bias
- 20 & 50 SMA not overextended from 200: only when both above 200; if $(SMA50 - SMA200) > 2 \times (SMA20 - SMA50)$ = overextended
- Price inside Keltner Channel (length 20, mult 1.25, ATR 14)
- Not between 3:30pm-6:02pm ET
- News blackout toggle off

Risk

- Stop Loss: 1 tick below lowest low of most recent bearish candle sequence before entry
- Take Profit: $ATR \times 12$ (14-period ATR on 1min, measured at candle before entry)
- Max SL filter: if SL distance $> ATR \times 5$, skip the trade

Moving Average Rules

MA order filter (1min and 15min):

- Block longs if: $SMA200 > SMA50 > SMA20$ (full bearish stack)
- Block shorts if: $SMA20 > SMA50 > SMA200$ (full bullish stack)
- Any mixed order = trade allowed

15min slope rule:

- For longs: at least one of 20 SMA or 50 SMA must be sloping upward or flat
- For shorts: at least one must be sloping downward or flat

1min overextension rule:

- Only applies when both SMA20 and SMA50 are above SMA200 (for longs)
- If $(SMA50 - SMA200) > 2 \times (SMA20 - SMA50)$ = overextended, no entry

Validation System (15min)

Type 1 validation:

- A bullish 3M wick candle prints after tapping the demand zone
- Bullish 3M candle: $low < low[1]$ AND $close \geq low[1]$
- For shorts: bearish 3M candle: $high > high[1]$ AND $close \leq high[1]$

Type 2 validation:

- A bullish engulfing candle after tapping the demand zone
- For shorts: bearish engulfing candle

15min Protected Low (15mpl)

- Type 1: 15mpl = bottom of the green 3M wick candle
- Type 2: 15mpl = bottom wick of candle immediately BEFORE the engulfing candle
- Updates to bottom of each swing candle
- Swing candle: 3M wick candle where right side never trades below the wick, and at some point continues higher
- Invalidation: candle BODY closes below 15mpl (wick below is fine)
- 15mpl must be set before any 1min entries are valid

Stage System (1min)

Finding a cluster (prerequisite for Stage 1):

- Identify the LOWEST unmitigated demand zone within the bullish range
- Price TAPS that zone (any wick into it)
- Price creates another mkU without a mkDn in between = CLUSTER = Stage 1 begins
- If mkDn happens between tap and next mkU = cluster search resets

Stage progression (longs):

Stage	Entry Rule	Condition
No Stage	No entries	Looking for cluster
Stage 1	Cluster breaks only (zone tapped first)	Breaks 2-4. nU resets on mkDn.
Stage 1 Reaccumulation	Structural or fractal entries	Model drops to Reaccumulation
Stage 2	Any mkU, breaks 1-4	Breaks up out of S1 Reaccumulation
Stage 2 Reaccumulation	Same as S1 Reaccumulation	mkDn in Stage 2
Late Stage 2	NO entries	Break out of S2 Reaccumulation + S2 had cluster
Reset/No Stage	No entries, seek cluster	Model turns bearish OR specific reset condition

Break numbering:

- Break 1: first mkU in range — not tradeable in Stage 1 (no cluster yet)
- Break 2: next mkU after zone tap = creates cluster = Stage 1 begins (tradeable if cluster)
- Breaks 3 and 4: tradeable in Stage 1 IF they are also clusters
- Stage 2: breaks 1-4 all tradeable (no cluster requirement)
- Stage 2 requires going through Stage 1 Reaccumulation first

Fractal Break System

Used during reaccumulation phases for fractal entries.

Finding the FB line:

- Find the LOWEST candle in the down move
- If lowest candle's high exceeds prior candle's high: draw horizontal line left from its high, find candle line touches = FB line goes on top of that candle
- Otherwise: FB line = HIGH of candle immediately BEFORE the lowest candle

Fractal entry sequence:

- FB1 zone must be tapped BEFORE FB2 breaks — else FB1 zone invalidated
- If FB1 zone tapped before FB2: buystop 1 tick above FB2 line
- FB2 zone must be tapped BEFORE FB3 breaks: buystop 1 tick above FB3 line
- Each new FB break creates fresh zone from new lowest candle
- Fractal reset: price breaks back below FB3 range = restart fractal count

- 2 resets in same reaccumulation phase = stop fractal entries until new structural break

Valid Zone Types for 15min Validation

- 1h structural zone (deepest unmitigated demand/supply in 1h range)
- 1h fractal zone
- 15min structural zone
- 15min fractal zone

Any one of these being tapped is sufficient to begin looking for Type 1 or Type 2 validation.

Special Setups

Double break reaccumulation:

- 2 consecutive structural breaks up (break 1 then break 2)
- Price breaks down into reaccumulation into unmitigated zone from break 1 = valid setup
- Same applies to fractals: FB1+FB2 then reaccumulation into FB1 zone

Automation Stack

TradingView (Pine Script alerts) → TradersPost (webhook) → Tradovate master demo → trade copier → prop firm accounts

Prop firm accounts:

- Apex Trader Funding, Lucid Trading, Tradify, MyFundedFutures, Alpha Futures
- Contracts: 1-60 micros depending on instrument

Current Code Status

File: 3M_ELITE_v2.pine

- Full indicator logic working (structure, models, MTF table)
- Zone detection functions `f_findDemandZone` and `f_findSupplyZone` — BEING DEBUGGED
- Debug labels showing `off=X max=Y` on each break

Known bug:

- For a bearish break, debug shows `off=3 max=29` but correct zone should be at offset 4
- Offset 4 has higher high, passes engulfing check, but something rejects it
- Suspected: mitigation check or one candle rule falsely rejecting offset 4

Correct zone function logic:

- Iterate every candle as potential ZONE CANDLE (not engulfing candle)
- Engulfing candle is at `cOff-1` (one bar to the right / closer to current bar)
- Demand: find zone candle with LOWEST LOW passing all checks

- Supply: find zone candle with HIGHEST HIGH passing all checks
- Engulfing: $\text{close}[\text{cOff}-1] > \text{high}[\text{cOff}]$ for demand; $\text{close}[\text{cOff}-1] < \text{low}[\text{cOff}]$ for supply
- Mitigation: candles from cOff-2 down to 1 — if $\text{low}[k] < \text{zone_high}$ AND $\text{high}[k] > \text{zone_low}$ = mitigated
- One candle rule: < 2 candles after engulfing trade beyond engulfing extreme